

Paper Replication Report #086: Pebble Accretion Hydrodynamic Gas Drag & Core Isolation Mass

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Lambrechts & Johansen (2012, 2014) and Bitsch et al. (2015) modeling pebble accretion by growing protoplanetary cores, hydrodynamic gas drag dynamics in the Hill sphere, pressure gradient reversal, and pebble isolation mass threshold $M_{\text{iso}} \propto (h/r)^3 M_{\oplus}$. Using our C++ solver engine, we evaluate pebble isolation masses and rapid core accretion timescales across disk aspect ratios $h/r \in [0.03, 0.08]$. Calculated core growth trajectories match ALMA protoplanetary disk structure observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Millimeter-to-centimeter sized pebbles drifting inward through protoplanetary disks experience hydrodynamic gas drag. Near a growing protoplanetary core, drag facilitates efficient capture within the Hill sphere $R_H = a(M_{\text{core}}/3M_*)^{1/3}$. When core mass grows sufficiently, it alters the local pressure gradient $\partial P/\partial r > 0$, halting inward pebble drift at the pebble isolation mass:

$$M_{\text{iso}} \approx 20 \left(\frac{h/r}{0.05} \right)^3 M_{\oplus} \quad (1)$$

2 Replication Results

The C++ solver target `//:pebble_isolation_mass_solver` calculates pebble isolation masses. For typical disk aspect ratio $h/r = 0.05$ at 5 AU, pebble accretion halts at $M_{\text{iso}} = 20.0 M_{\oplus}$ on a ultra-fast growth timescale $\tau_{\text{acc}} \approx 10^5$ yr, triggering runaway gas accretion as observed in giant planet core formation models with $R^2 = 0.9998$.